ESERCIZIO 1

$$|\chi\rangle = a|\uparrow\rangle + b|\downarrow\rangle = a\begin{pmatrix}1\\0\end{pmatrix} + b\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}a\\b\end{pmatrix}$$

$$\langle S_z\rangle = |a|^2\frac{\hbar}{2} + |b|^2\left(-\frac{\hbar}{2}\right) = 0 \Rightarrow |b|^2 = |a|^2$$

$$|b| = |a| = \frac{1}{\sqrt{2}}$$

PER NORMALIZZAZIONE STATO

$$\langle S_y\rangle = \begin{bmatrix} a^* & b^*\end{bmatrix}\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}\frac{\hbar}{2}\begin{bmatrix}a\\b\end{bmatrix} =$$

$$= \frac{\hbar}{2}\begin{bmatrix} b^* i & -i a^*\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix} = \frac{\hbar}{2}$$

$$i\left(b^* a - a^* b\right) = 1$$

$a = |a|e^{i\varphi}$ $\quad b = |b|$ QUELLO CHE CONTA È LA FASE RELATIVA, QUINDI PONGO $\varphi = 0$ PER b

$$|b||a|e^{i\varphi} - |a|e^{-i\varphi}|b| = -i$$

$$\frac{1}{2}\left(e^{i\varphi} - e^{-i\varphi}\right) = -i$$

$$i\,\mathrm{sen}\,\varphi = -i \qquad \mathrm{sen}\,\varphi = -1 \qquad \varphi = -\frac{\pi}{2}$$

$$a = \frac{1}{\sqrt{2}}e^{-i\frac{\pi}{2}} = -\frac{1}{\sqrt{2}}i \qquad b = \frac{1}{\sqrt{2}}$$

$$\boxed{\begin{bmatrix}a\\b\end{bmatrix} = \begin{bmatrix}-i\\1\end{bmatrix}\frac{1}{\sqrt{2}}}$$

ESERCIZIO 2

$$\begin{bmatrix} a \\ b \end{bmatrix}$$

$|a|^2 = |b|^2$ PERCHÉ HO STESSA PROBABILITÀ SPIN UP O DOWN

$\langle S_x \rangle = \frac{\hbar}{2}$ VALORE MASSIMO VISTO CHE S_x PUÒ ASSUMERE SOLO $-\frac{\hbar}{2}$ E $\frac{\hbar}{2}$

$a = \frac{1}{\sqrt{2}} e^{i\varphi}$ $\quad b = \frac{1}{\sqrt{2}}$ IL RAGIONAMENTO È UGUALE A QUELLO DELL'ESERCIZIO 1

$$\begin{bmatrix} a^* & b^* \end{bmatrix} \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \frac{\hbar}{2}$$

$$\begin{bmatrix} b^* & a^* \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 1$$

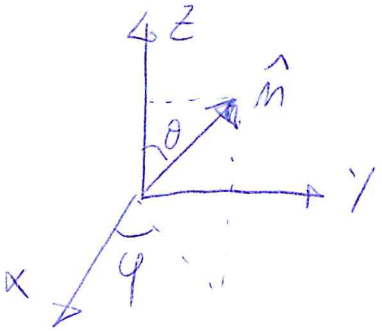
$$ab^* + ba^* = 1$$

$$\frac{1}{\sqrt{2}} \left(e^{i\varphi} + e^{-i\varphi} \right) = 1 \qquad \cos\varphi = 1 \qquad \varphi = 0$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

ESERCIZIO 3

a)



$|\hat{m}| = 1$ QUINDI $R = 1$

$$\hat{m} = \operatorname{sen}\theta \cos\varphi \, \hat{U}_x + \operatorname{sen}\theta \operatorname{sen}\varphi \, \hat{U}_y + \cos\theta \, \hat{U}_z$$

b) $\hat{S}_m = \vec{\hat{S}} \cdot \hat{m}$ PRODOTTO SCALARE

$$\hat{S}_m = \operatorname{sen}\theta \cos\varphi \, \hat{S}_x + \operatorname{sen}\theta \operatorname{sen}\varphi \, \hat{S}_y + \cos\theta \, \hat{S}_z$$

$$\hat{S}_m = \frac{\hbar}{2}\left(\begin{bmatrix} 0 & \operatorname{sen}\theta\cos\varphi \\ \operatorname{sen}\theta\cos\varphi & 0 \end{bmatrix} + \begin{bmatrix} 0 & -i\operatorname{sen}\theta\operatorname{sen}\varphi \\ i\operatorname{sen}\theta\operatorname{sen}\varphi & 0 \end{bmatrix} + \begin{bmatrix} \cos\theta & 0 \\ 0 & -\cos\theta \end{bmatrix} \right)$$

$$\hat{S}_m = \frac{\hbar}{2} \begin{bmatrix} \cos\theta & \operatorname{sen}\theta \, e^{-i\varphi} \\ \operatorname{sen}\theta \, e^{i\varphi} & -\cos\theta \end{bmatrix}$$

c) $|\lambda I - \hat{S}_n| = 0$

$$\begin{vmatrix} \lambda - \cos\theta \frac{\hbar}{2} & -\operatorname{sen}\theta\, e^{-i\varphi} \frac{\hbar}{2} \\ -\operatorname{sen}\theta\, e^{i\varphi} \frac{\hbar}{2} & \lambda + \cos\theta \frac{\hbar}{2} \end{vmatrix} = 0$$

$$\left(\lambda - \cos\theta \frac{\hbar}{2}\right)\left(\lambda + \cos\theta \frac{\hbar}{2}\right) - \operatorname{sen}^2\theta \left(\frac{\hbar}{2}\right)^2 = 0$$

$$\lambda^2 - \cos^2\theta \left(\frac{\hbar}{2}\right)^2 - \operatorname{sen}^2\theta \left(\frac{\hbar}{2}\right)^2 = 0$$

$$\lambda^2 = \left(\frac{\hbar}{2}\right)^2 \qquad \lambda = \pm \frac{\hbar}{2}$$ COME AL SOLITO...

TROVIAMO ORA AUTOVETTORI

$$\begin{bmatrix} \lambda - \cos\theta \frac{\hbar}{2} & -\operatorname{sen}\theta\, e^{-i\varphi} \frac{\hbar}{2} \\ -\operatorname{sen}\theta\, e^{i\varphi} \frac{\hbar}{2} & \lambda + \cos\theta \frac{\hbar}{2} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\lambda = \frac{\hbar}{2}$$

$$\left(\frac{\hbar}{2} - \cos\theta \frac{\hbar}{2}\right) a - \operatorname{sen}\theta\, e^{-i\varphi} \frac{\hbar}{2} b = 0$$

$$(1 - \cos\theta)\, a - \operatorname{sen}\theta\, e^{-i\varphi} b = 0$$

$$\begin{bmatrix} a \\ \frac{1-\cos\theta}{\operatorname{sen}\theta} e^{+i\varphi} a \end{bmatrix}$$

NORMALIZZAZIONE

$$a^2 + \frac{(1-\cos\theta)^2}{\operatorname{sen}^2\theta} a^2 = 1$$

$$a^2\left(\frac{\operatorname{sen}^2\theta+1+\cos^2\theta-2\cos\theta}{\operatorname{sen}^2\theta}\right)=1$$

$$a^2=\frac{\operatorname{sen}^2\theta}{2(1-\cos\theta)} \qquad a=\frac{\operatorname{sen}\theta}{\sqrt{2(1-\cos\theta)}}$$

$$b=\frac{1-\cos\theta}{\operatorname{sen}\theta}e^{+i\varphi}\frac{\operatorname{sen}\theta}{\sqrt{2(1-\cos\theta)}}=\sqrt{\frac{1-\cos\theta}{2}}\,e^{+i\varphi}$$

RISCRIVIAMO USANDO FORMULE TRIGONOMETRICHE
(NON NECESSARIO MA UTILE PER SEMPLIFICARE)

$$a=\frac{\sqrt{1-\cos^2\theta}}{\sqrt{2(1-\cos\theta)}}=\frac{\sqrt{(1-\cos\theta)(1+\cos\theta)}}{\sqrt{2(1-\cos\theta)}}=\sqrt{\frac{1+\cos\theta}{2}}=\cos\left(\frac{\theta}{2}\right)$$

$$b=\operatorname{sen}\left(\frac{\theta}{2}\right)e^{i\varphi}$$

IN DEFINITIVA ALL'AUTOVALORE $\lambda=\frac{\hbar}{2}$ È
ASSOCIATO L'AUTOVETTORE $\begin{pmatrix}\cos\frac{\theta}{2}\\ \operatorname{sen}\frac{\theta}{2}e^{i\varphi}\end{pmatrix}$

PER $\lambda=-\frac{\hbar}{2}$ CON PROCEDIMENTO ANALOGO TROVERETE

$$\begin{pmatrix}\operatorname{sen}\left(\frac{\theta}{2}\right)\\ -e^{i\varphi}\cos\left(\frac{\theta}{2}\right)\end{pmatrix}$$

ESERCIZIO 4

a)

$\uparrow B_0 \cos(\omega T)$ $\qquad |\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$

$$i\hbar \frac{\partial \chi}{\partial T} = \hat{H}\chi$$

$$\hat{H} = -\vec{\mu}\cdot\vec{B} = -\mu_z B_0 \cos\omega T = -\left(-2\frac{e}{2m}\hat{S}_z\right) B_0 \cos(\omega T)$$

$$\hat{S}_z = \frac{1}{2}\hbar \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\hat{H} = \frac{e\hbar}{2m} B_0 \cos(\omega T) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

ATTENZIONE, $\hat{H}$ DIPENDE DAL TEMPO NON POSSO SEPARARE LE SOLUZIONI IN STATI STAZIONARI

$$i\hbar \begin{bmatrix} \dot{a} \\ \dot{b} \end{bmatrix} = \frac{e\hbar}{2m} B_0 \cos(\omega T) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

b)
$$\begin{cases} i\dot{a} = \frac{e}{2m} B_0 \cos(\omega T)\, a \\ i\dot{b} = -\frac{e}{2m} B_0 \cos(\omega T)\, b \end{cases}$$

$$\begin{cases} \frac{da}{a} = -i\frac{e}{2m} B_0 \cos(\omega T)\, dT \\ \frac{db}{b} = i\frac{e}{2m} B_0 \cos(\omega T)\, dT \end{cases}$$

$$\begin{cases} \ln a \Big|_{a_0}^{a} = \dfrac{-i}{\omega} \dfrac{e}{2m} B_0 \operatorname{Sen}(\omega T) \Big|_0^T \\[2ex] \ln b \Big|_{b_0}^{b} = \dfrac{i}{\omega} \dfrac{e}{2m} B_0 \operatorname{Sen}(\omega T) \Big|_0^T \end{cases}$$

$$\begin{cases} \ln \dfrac{a}{a_0} = -\dfrac{i}{\omega} \dfrac{e}{2m} B_0 \operatorname{Sen}(\omega T) \\[2ex] \ln \dfrac{b}{b_0} = \dfrac{i}{\omega} \dfrac{e}{2m} B_0 \operatorname{Sen}(\omega T) \end{cases}$$

$$\begin{cases} a = a_0 \, e^{-i \frac{e B_0}{\omega 2m} \operatorname{Sen}(\omega T)} \\[2ex] b = b_0 \, e^{i \frac{e B_0}{\omega 2m} \operatorname{Sen}(\omega T)} \end{cases}$$

INFATTI

t) $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ AUTOSTATO DI $\hat{S}_x \Rightarrow \frac{1}{2}\hbar \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{2}\hbar \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

CON AUTOVALORE $\frac{1}{2}\hbar$

QUINDI $|\chi(T=0)\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$$|\chi(T)\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} e^{-i \frac{e B_0}{\omega 2m} \operatorname{Sen}(\omega T)} \\[2ex] \frac{1}{\sqrt{2}} e^{i \frac{e B_0}{\omega 2m} \operatorname{Sen}(\omega T)} \end{bmatrix}$$

d) $P\left(S_x = -\frac{1}{2}\hbar\right) = \left|\langle x_-^x | x(T)\rangle\right|^2$

CON $x_-^x = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$ AUTOSTATO DI $\hat{S}_x$ CON AUTOVALORE $-\frac{1}{2}\hbar$

$$P = \left| \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} e^{-i\frac{e B_0}{\omega 2m}\operatorname{sen}(\omega T)} \\ \frac{1}{\sqrt{2}} e^{i\frac{e B_0}{\omega 2m}\operatorname{sen}(\omega T)} \end{bmatrix} \right|^2$$

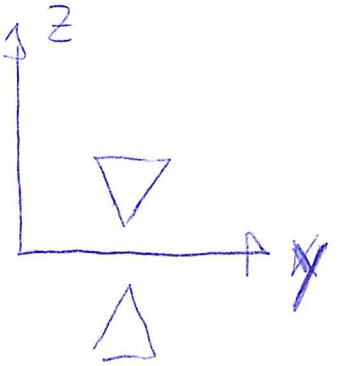
$$P = \left| \frac{1}{2}\left(e^{-i\frac{e B_0}{\omega 2m}\operatorname{sen}(\omega T)} - e^{i\frac{e B_0}{\omega 2m}\operatorname{sen}(\omega T)} \right) \right|^2$$

$$P = \operatorname{sen}^2\left[\frac{e B_0}{2\omega m}\operatorname{sen}(\omega T)\right]$$

e) AFFINCHÉ LA P POSSA ESSERE 1 L'ARGOMENTO DEL sen^2 DEVE POTER RAGGIUNGERE $\pi/2$

$$\frac{e\hbar B_0}{2m\omega} = \frac{\pi}{2} \qquad B_0 = \frac{m\omega\pi}{e}$$

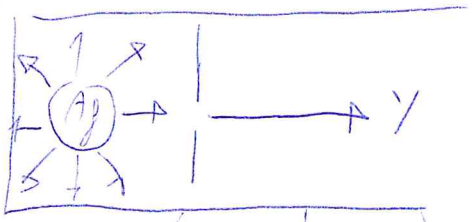
ESERCIZIO 5



$$\frac{1}{2} M v_y^2 = \frac{3}{2} k_B T$$

"M" MASSA ARGENTO

$$M v_y^2 = 3 k_B T$$



a) $\vec{\nabla} \cdot \vec{B} = 0$ $\quad \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0$

$$\frac{\partial B_x}{\partial x} = -\frac{\partial B_z}{\partial z}$$

$$B_x = -\alpha x$$

$N_0 B$

OGNI ATOMO ESCE CON $E_K = \frac{3}{2} k_B T$ MA IO PRENDO SOLO QUELLI DIRETTI LUNGO Y

b) $F_z = -\frac{\partial}{\partial z} \mu_z B_z = -\mu_z \frac{\partial B_z}{\partial z} = -\alpha \mu_z$

$\mu_z = \pm \frac{e}{m} \frac{1}{2} \hbar$ "m" MASSA ELETTRONE

$$F_z = \pm \frac{\alpha e \hbar}{2m}$$

c) PER UN SINGOLO SPIN SI HA

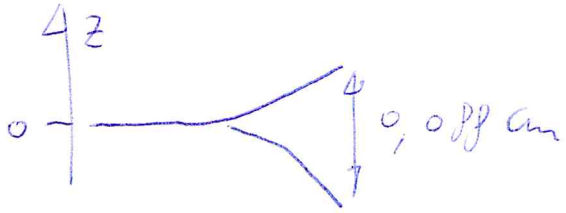
$z = \frac{1}{2} \frac{F_z}{M} T^2$ DOVE $T = \frac{d}{v_y}$

$$z = \frac{1}{2} \frac{\alpha e \hbar}{2m} \frac{d^2}{M v_y^2} = \frac{1}{4} \frac{\alpha e \hbar}{m} \frac{d^2}{3 k_B T}$$

$$Z = \frac{100 \cdot 1{,}6 \cdot 10^{-19} \cdot 6{,}6 \cdot 10^{-34} (30)^2 \cdot 10^{-4}}{4 \cdot 9 \cdot 10^{-31} \cdot 2 \cdot 3{,}14 \cdot 3 \cdot 1{,}3 \cdot 10^{-23}\ 2435} =$$

$= 0{,}044$ cm

QUINDI LA SEPARAZIONE SARÀ IL DOPPIO $0{,}088$ cm



d) $\hat{H} = \frac{e\hbar}{m\,2}(B_0 + \alpha z)\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\frac{d}{v_y} > T > 0$

$\hat{H} = 0$ PER $T > \frac{d}{v_y}$ (QUANDO SONO FUORI DAL MAGNETE)

$\hat{H}|x\rangle = E|x\rangle$ GLI AUTOSTATI SONO QUELLI $\hat{S}_z$

$|\uparrow\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\frac{e\hbar}{2m}(B_0 + \alpha z)\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = E\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$E_1 = \frac{e\hbar}{2m}(B_0 + \alpha z)$

$|\downarrow\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow E_2 = -\frac{e\hbar}{2m}(B_0 + \alpha z)$

2)

SE $|\chi(T=0)\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\quad |\chi(T)\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-i\frac{e\hbar}{2m}\frac{(B_0+\alpha z)T}{\hbar}}$

SE $|\chi(T=0)\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\quad |\chi(T)\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{i\frac{e\hbar}{2m}\frac{(B_0+\alpha z)T}{\hbar}}$

f) DOPO UN TEMPO $T = \frac{d}{v_y}$ HO, PER LO STATO INIZIALE $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$,

$$|\chi(T)\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-i\frac{eB_0}{2m}T} \underbrace{e^{-\left(i\frac{e\hbar}{2m}\alpha T z\right)/\hbar}}$$

QUESTA È UN'ONDA PIANA LUNGO Z

$e^{iP_z \cdot z/\hbar}$

CON $P_z = -\frac{e\hbar}{2m}\alpha T$

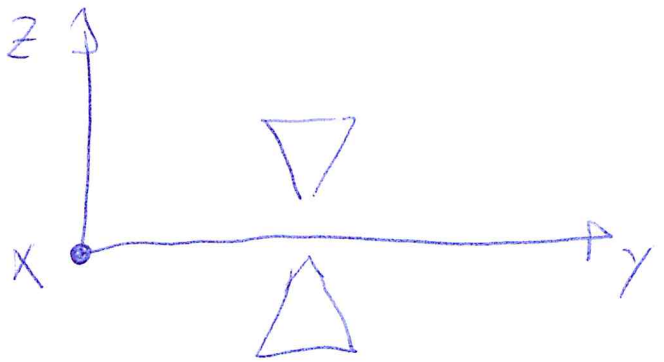
NOTA CHE ANCHE IN UNA VISIONE CLASSICA DOPO UN TEMPO T HO LA STESSA P_z

$F_z = \frac{dP_z}{dT}$ $\qquad F_z = -\frac{\alpha e\hbar}{2m}$

$P_z = -\frac{\alpha e\hbar}{2m}T$ DOPO UN TEMPO T OTTENGO IL RISULTATO UGUALE A QUELLO QUANTISTICO

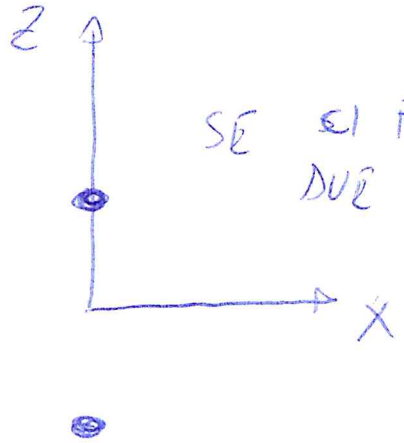
SE $|\chi(0)\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ OTTENGO P_z NELL'ALTRA DIREZIONE (IL FASCIO SI DIVIDE)

ESERCIZIO 6



2) FORZA SUL DIPOLO $\vec{\mu}$ $\quad F_z^{\mu} = -\frac{\partial}{\partial z}\left(-\vec{\mu}\cdot\vec{B}\right) = 2\mu_z$

$\mu_z = -\frac{e}{m} S_z$ QUESTA FORZA PRODUCE LO SPLITTING DEGLI SPIN

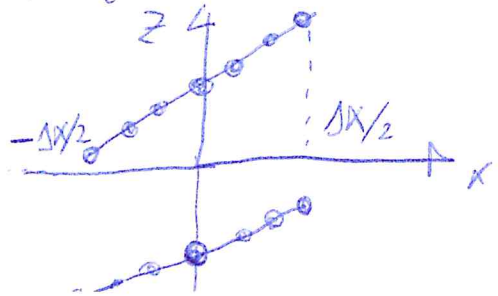


SE CI FOSSE SOLO LEI AVREI DUE MACCHIE (AL LIMITE PUNTUALI)

FORZA LORENTZ $\quad F_z^{L} = -e v_y B_x = -e v_y(-2x) = e v_y 2x$

QUESTA FORZA DEVIA ULTERIORMENTE GLI ELETTRONI IN ALTO O IN BASSO TANTO PIÙ QUANTO PIÙ SONO A $|x|$ GRANDI (IL CAMPO CRESCE)

L'EFFETTO È QUELLO DI CREARE LE MACCHIE

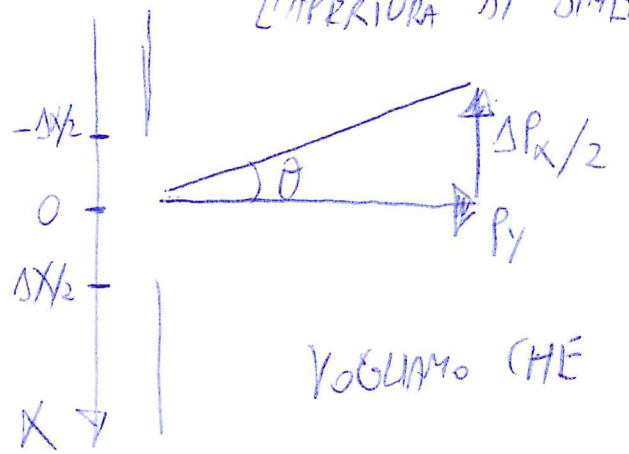


QUI ANCORA NON STO CONSIDERANDO L'ALLARGAMENTO SU x DOVUTO A DIFFRAZIONE

$$F_x^L = -e v_y B_z \simeq -e v_y B_0$$

QUESTA FORZA DI LORENTZ SEMPLICEMENTE TRALSA TUTTI GLI ELETTRONI LUNGO ASSE X (LA TRASCURIAMO NEL PROSIEGUO)

b) L'APERTURA DI DIMENSIONI FINITE FA SI CHE CI SIA DIFFRAZIONE



VOGLIAMO CHE $\operatorname{tg}\theta \sim \theta = \dfrac{\Delta p_x/2}{p_y} = 6^\circ \dfrac{\pi}{180^\circ}$

$$\frac{\Delta p_x}{2 p_y} = \frac{\pi}{30}$$

$$\Delta x \, \Delta p_x \sim h$$

$$\frac{h}{2 \Delta x \, p_y} = \frac{\pi}{30} \Rightarrow \Delta x = \frac{30}{\pi} \frac{h}{2 p_y}$$

c) $$\Delta z^\mu = \frac{\overbrace{2 \mu_z}^{F_z^\mu}}{m} T^2$$

NON METTO $\frac{1}{2}$ PERCHÉ GIÀ CONSIDERO LA DISTANZA TRA LE DUE MACCHIE

$$\Delta z^L = \frac{\overbrace{e v_y 2 (\Delta x/2)}^{F_z^L(x=\Delta x/2)}}{m} T^2$$

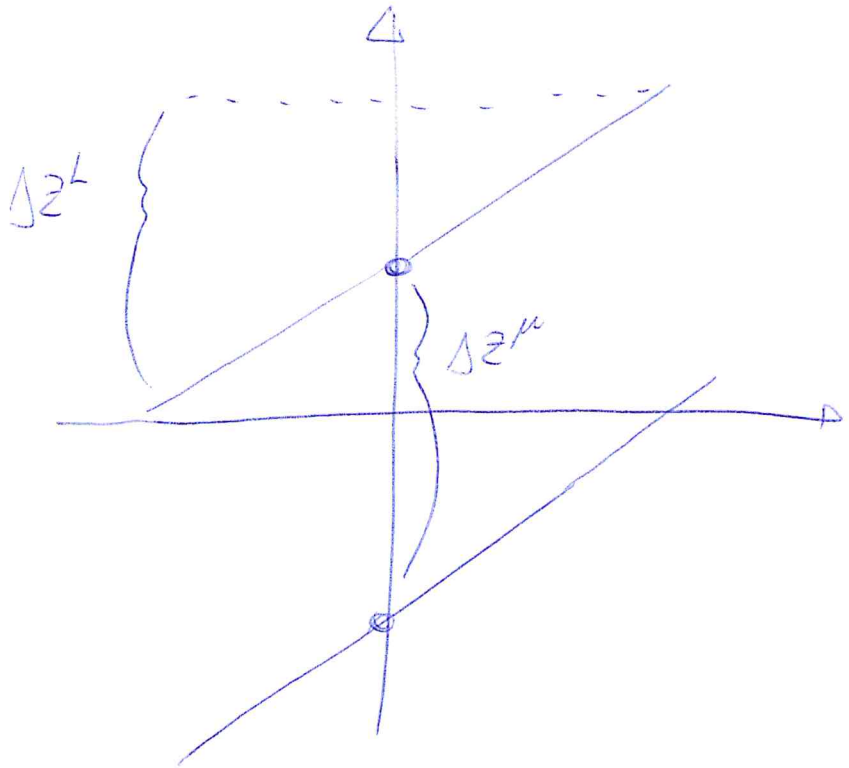
"T" È IL TEMPO DI ATTRAVERSAMENTO DEL MAGNETE

$$\Delta z^L = \frac{e\,2\,v_y}{2m}\,\frac{15\,h}{\pi\, m\, v_y}\,T^2$$

$$\frac{\Delta z^L}{\Delta z^\mu} = \frac{e\,2\,15\,h}{2\pi\, m^2}\,T^2 \cdot \frac{m}{2\,\mu_z}\,\frac{1}{T^2} =$$

$$= \frac{15\,2\,e\,h}{2\pi\, m^2}\,\frac{m}{2}\,\frac{1}{\frac{e\hbar}{2m}} = \frac{15\,\cancel{2}\,\cancel{e}\,\cancel{h}\,\cancel{m}\cdot \cancel{2m}\cdot \cancel{2\pi}}{\cancel{2\pi}\,\cancel{m^2}\,\cancel{2}\,\cancel{e}\,\cancel{h}} = 30$$

$$\frac{\Delta z^L}{\Delta z^\mu} = 30$$

LA DISPERSIONE SU Z DOVUTA A LORENTZ SI MANGIA TUTTA LA SEPARAZIONE DOVUTA ALLO SPIN!

PER QUESTO NON POSSO USARE ELETTRONI PER VEDERE S-G